La Consolacion University Philippines
Valenzuela St., Capitol View Park Subdivision, Bulihan, City of Malolos, Bulacan

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SCHOOL YEAR 2025-2026
COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. COURSE CODE : COE7

II. COURSE TITLE : Computer Engineering Drafting and Design

III. COURSE DESCRIPTION :

The course covers the basic AutoCAD commands and computer-aided drafting concepts, to draw, design and draft. Emphasis is placed on efficient and accurate drawing techniques incorporating the features, commands, and techniques for creating, editing, and printing 3D production drawings. Students will also learn how to design through hands-on laboratory sessions and course projects. Values are also integrated in each learning content to ensure not only intellectual development of the students but to develop Christo-centric Catholic engineers.

IV. PRE-REQUISITE : CAD

I.V. CREDIT UNITS : 1 Unit Laboratory 3 Units Lecture

VI. COURSE LEARNING OUTCOMES:

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development of Catholic Augustinian Marian graduates rooted in Gospel values.	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them: ● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;*

	● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* ● *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Ability to apply knowledge of mathematics and science to solve complex engineering problems; PO2. Ability to design and conduct experiments, as well as to analyze and interpret data; PO3. Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability, in accordance with standards. PO4. Ability to function on multidisciplinary teams; PO5. Ability to identify; formulate, and solve complex engineering problems; PO6. Understanding of professional and ethical responsibility. PO7. Ability to communicate effectively; PO8. Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.	CLO1: Identify the fundamentals in preparing technical drawings and design schematics for computer and electronic systems. [EGA 1] [PO1, PO2, PO3, PO5] CLO2: Design and develop system layouts and components using engineering drawing standards and CAD tools, considering manufacturability, sustainability, and safety. [EGA 2] [PO 4, 7, 12] CLO 3: Collaborate effectively within multidisciplinary teams to design and present engineering projects using clear technical communication. [EGA3] [PO4, PO7, PO12] CLO3: Demonstrate proficiency in using modern drafting and modeling software tools for circuit, system, and architectural design applications. [EGA 4] [PO11, PO13]

	PO9. Recognition of the need for, and an ability to engage in life-long learning; PO10. Knowledge of contemporary issues; PO11. Ability to use techniques, skills and modern engineering tools necessary for engineering practice; and PO12. Knowledge and understanding of engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments. PO13: Ability to design, develop, implement and improve integrated system that include people materials, implementation, equipment and energy.	

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENT	SUGGESTED TEACHING-LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUES TO BE INTEGRATED
Week 1	LO7: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing.	**1. Orientation** **A.** Introduction to the Course Contents **B.** Course Requirements and Activities **C.** Course Objectives **D.** Grading System and Policies on	• Setting and sharing of expectations from the course • Discussion of the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course	• Statement of expectations • Review of Course Contents, Objectives, Activities and Assessment Tools Review of Policies on Absences, Tardiness,	• Understanding • Open Mindedness • Respect Obedience

		Absences, Tardiness and Examinations	Requirements and Activities Review of Policies on Absences, Tardiness, Examinations and Grading System	Examinations and Grading System	
Week 2	ILO1: Navigate the AutoCAD user interface. [CLO1] ILO2: Show one's knowledge and understanding on AutoCAD's fundamental features. [CLO1]	**2. Creating 2D Drawing** • Recap on creating basic drawing • Line • Polyline • Circle • Arc • Rectangle Ellipse	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness
Week 3 – Week 4	ILO1: Show one's knowledge and understanding on using the precision drafting tools to develop accurate technical drawings. [CLO1] ILO2: Present the drawing in a detailed and visually impressive manner. [CLO2]	• Spline • Helix • Donut Polygon • Hatch • Move • Copy • Stretch • Rotate • Mirror • Scale • Trim • Fillet • Array • Offset Explode	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness

	ILO3: Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [CLO2]				
Week 5	ILO3: Show one's knowledge and understanding on using the precision drafting tools to develop accurate technical drawings. [CLO2] ILO4: Present the drawing in a detailed and visually impressive manner. [CLO2]	**3. Creating 3D Drawing** • 3D Drawing • Box • Cylinder • Cone • Sphere • Pyramid • Wedge • Torus Polysolid	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness
Week 6		**Preliminary Examination**	Examination on the topics discussed during the Preliminary Term	Examination	• Honesty in answering the test questionnaire • Diligence • Understanding

Week 7	ILO1: Apply design principles to convert 2D sketches into accurate 3D models using CAD tools. [CLO1] ILO2: Use modern CAD software techniques to create and modify 3D models for engineering applications. [CLO2]	• 2D to 3D Drawing • Extrude • Loft • Revolve • Sweep Presspull	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement Open mindedness
Week 8 – Week 10	**ILO1:** Analyze 3D designs to ensure accuracy, functionality, and compliance with engineering standards. [CLO3] **ILO 4:** Communicate and present 3D design outputs effectively	**4. Editing 3D Drawing** • Union • Subtract • Intersecting • Slice • Thicken • Separate • Shell • Edit Faces • Extrude • Taper • Move • Copy • Offset	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness

	through technical documentation and visualization. visually impressive manner. [CLO4]	• Edit Edges • Extract • Color • Copy • 3D Mirror • 3D Rotate • 3D Move 3D Scale			
Week 11	ILO1: Demonstrate the ability to navigate and manipulate 3D environments to effectively visualize and analyze engineering designs. [CLO2\]	**5. Visualizing 3D Drawing** • Coordinate Visual Styles	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness
Week 12	CLO1-CLO4	**Midterm Examination**	Examination on the topics discussed during the Mid Term	Examination	• Honesty in answering the test questionnaire • Diligence • Understanding
Week 13 – Week 14	ILO1: Identify and apply different coordinate systems used in 3D modeling for accurate object	**6. 3D Drawing Mesh** • Mesh Box • Mesh Cylinder • Mesh Pyramid	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory	• Cooperation • Understanding Critical Thinking

	placement and visualization. [CLO2]	• Mesh Sphere • Mesh Wedge • Mesh Torus • Smoothing • Creasing • Converting to Solid Converting to Surface			
Week 15 – Week 16	ILO 1: Create and manipulate various mesh objects such as box, cylinder, pyramid, sphere, wedge, and torus to form complex 3D models. [CLO2-3] **ILO 2:** Apply mesh editing techniques such as smoothing and creasing to refine 3D designs. [CLO3-4] **ILO 3:** Convert mesh models into solid and surface forms to prepare them for engineering	**7. 3D Rendering** • Lighting • Sun • Sky • Materials	• Lecture-Discussion using Power Point presentation	• Quiz on the topics discussed Laboratory Activities	• Cooperation • Understanding • Critical Thinking • Active participation and involvement • Open mindedness

	visualization and analysis. [CLO3-4]				
Week 17	[CLO1-CLO4]	**Final Examination**	Examination on the topics discussed during the Final Term	• Examination	• Honesty in answering the test questionnaire • Diligence • Understanding

VIII. COURSE REQUIREMENTS :

1. Class Attendance

2. Recitations

3. Quizzes

4. Examinations

5. Compilation of Laboratory Activity Reports

6. Project

IX. REFERENCES:

Yarwood, A. (2013). Introduction to AutoCAD 2013: 2D and 3D Design. New York. United States of America: Routledge ISBN: 978-0415537629

Holland, L. (2012). *Mastering AutoCAD Civil 3D 2013.* United States of America: John Wiley & Sons ISBN: 978-1118281758

Onstott, S. (2012). *AutoCAD 2013 and AutoCAD LT 2013.* Indianapolis. United States of America: John Wiley & Sons ISBN: 978-

X. GRADING SYSTEM :

1. Class Standing 60%
   - Quizzes / Laboratory Activities 30%
   - Project 30%
   - Recitation 30%
   - Attitude/Attendance 10%
2. Term Examination 40%
3. Term Grade = Class Standing (60%) + Term Examination (40%)
4. Final Rating = $\frac{\text{Preliminary Term Grade + Mid Term Grade + Final Term Grade}}{3}$

Prepared by:	Checked and Reviewed by:	Approved by:
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Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	